

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PNG 310 — Past Questions & Materials
by Renance

Course: PNG 310
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 300, 400

This study pack was compiled and restructured by Renance for PNG 310 students at Federal University of Petroleum Resources, Effurun. It brings together 5 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 5 — PNG 310 2023/2024 Past Question

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY
Department of Petroleum Engineering

2023/2024 First Semester Examination

PNG 310: Introduction to Petroleum Engineering (for non-Petroleum Students)

Department of Computer Science Students Only

Answer Question 1 and any other three.

Time Allowed: 2Hrs: 30mins.

QUESTIONS

- 1(a)i. Explain the difference between abiogenic and biogenic origins of petroleum.
ii. Describe the process of hydrocarbon migration.
- (b)i. What role does the source rock play in the formation of petroleum?
ii. Calculate the API gravity of a crude oil sample that has a specific gravity of 0.85. State the crude classification based on the calculated API.
- (c)i. Describe the role of development wells in the oil production phase and their impact on the field's productivity.
ii. What is the primary objective of production operations in the oil and gas industry?
- 2(a)i. What is a petroleum trap, and why is it significant?
ii. Define API gravity and its significance in petroleum engineering.
- (b)i. Differentiate between solution gas and free gas in a reservoir.
ii. Identify and describe the three main reservoir drive mechanisms.
- 3(a)i. Explain Enhanced Oil Recovery (EOR).
ii. Define porosity and its importance in petroleum engineering.
- (b)i. Explain the concept of permeability in the context of petroleum engineering.
ii. What is the role of tubing in a completed oil or gas well?
- 4(a)i. Define "exploration well" and discuss its primary objective in the oil exploration process.
ii. Explain the significance of appraisal wells in the evaluation of a newly discovered oil field.
- (b)i. Explain what a relief well is and its purpose in managing blowouts.
ii. Define Drilling Engineering.
- 5(a)i. What does it mean when a well is considered "dry"?
ii. List three types of casings used in well completion and their purposes.
- (b)i. Explain the purpose of perforating the casing in a completed well.
ii. When is artificial lift applied in oil and gas production?

Paper 2 of 5 — PNG 310 APPLIED RESERVOIR ENGINEERING 2024/2025

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

COLLEGE OF TECHNOLOGY

DEPARTMENT OF PETROLEUM ENGINEERING ult of a gas chromatographic test performe

FIRST SEMESTER EXAMINATION, 2024-2025 SESSION as sample collected from © gas field in the Niger Delta region Is

. een a ae ee. See in the tabie below

PNG APPLIED RESERVOIR ENGINEERING —

Date: 13-04-2025 TIME ALLOWED: 3HR UNIT: 3 Cou _~onents Mole Percent (x)

Nitrogen 0.18

INSTRUCTION: ANSWER THREE QUESTION IN SESSION A AND Carbon dioxide 0.91

TWO IN SESSION B | Methane 87.43

Ethane 441

SESSION A Propane 3.65

QUESTION 1 i-butane 0 64

; n-butane 1.50

An active water drive reservoir has 2 cumulative water production of 28640 i-pentane 0.45 =

bbl was estimated after the first year of production The pressure profile, ~n-pentane 0 38 -

reservoir and aquifer properties are given as (Hex ae 0 30

Time (yrs) —T0 a 5 | — 7 [5 5 | | Heptane Plus 015 ei

Pressure (psia) | 6027 5016 | 5014 | 4966 “\aete | 4775 B.) information:

Re Wen 5 3168 | 8.7389 | 11.7766 14 6216 | 17.3312 | ‘he formation:

Aquifer permeability, Kn 760 mD initial reservoir pressure, Py 3200 psia

Water viscosity, Hw 0.78 cp Reservoir thickness, N 55 ft

Drainage radius, Te 9500 tt Connas. water sats ation, Sw< 23%

Aquifer radius, Ta 47500 ft arpa, A

Total compressibility, C; 380° 10° psi' ; 10 acres

Porosily, ® 22% Porosity, 9 19.8%

Aquifer angle __ 120" Reservoir temperature _ 200°F

i Calculate the cumulative gas produced at 2500 psia

ative water influx for each year

ji, Calculate the recovery factor at 2500psia

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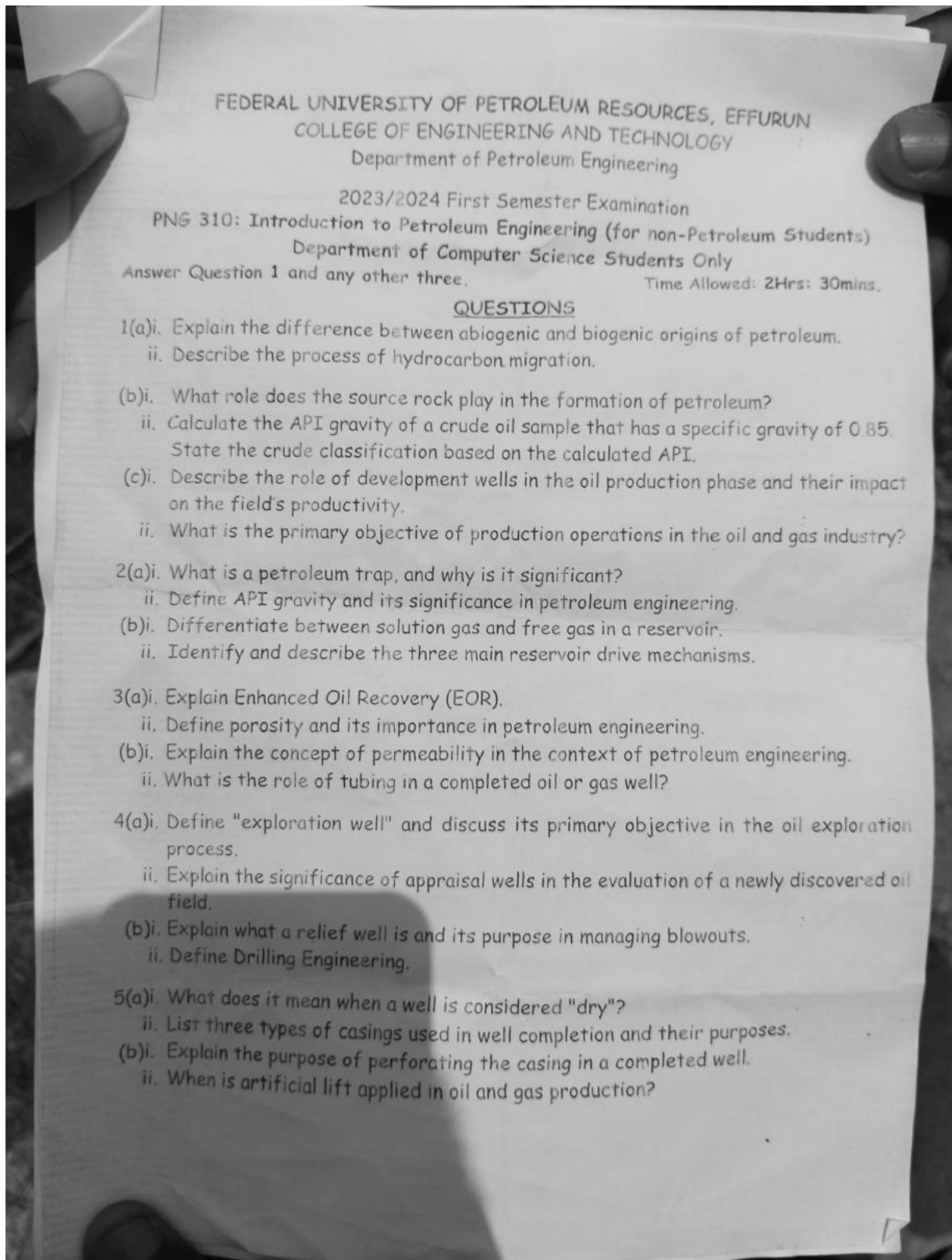
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Paper 3 of 5 — PNG 310 PQ

Level 400 • First Semester • PastQuestion

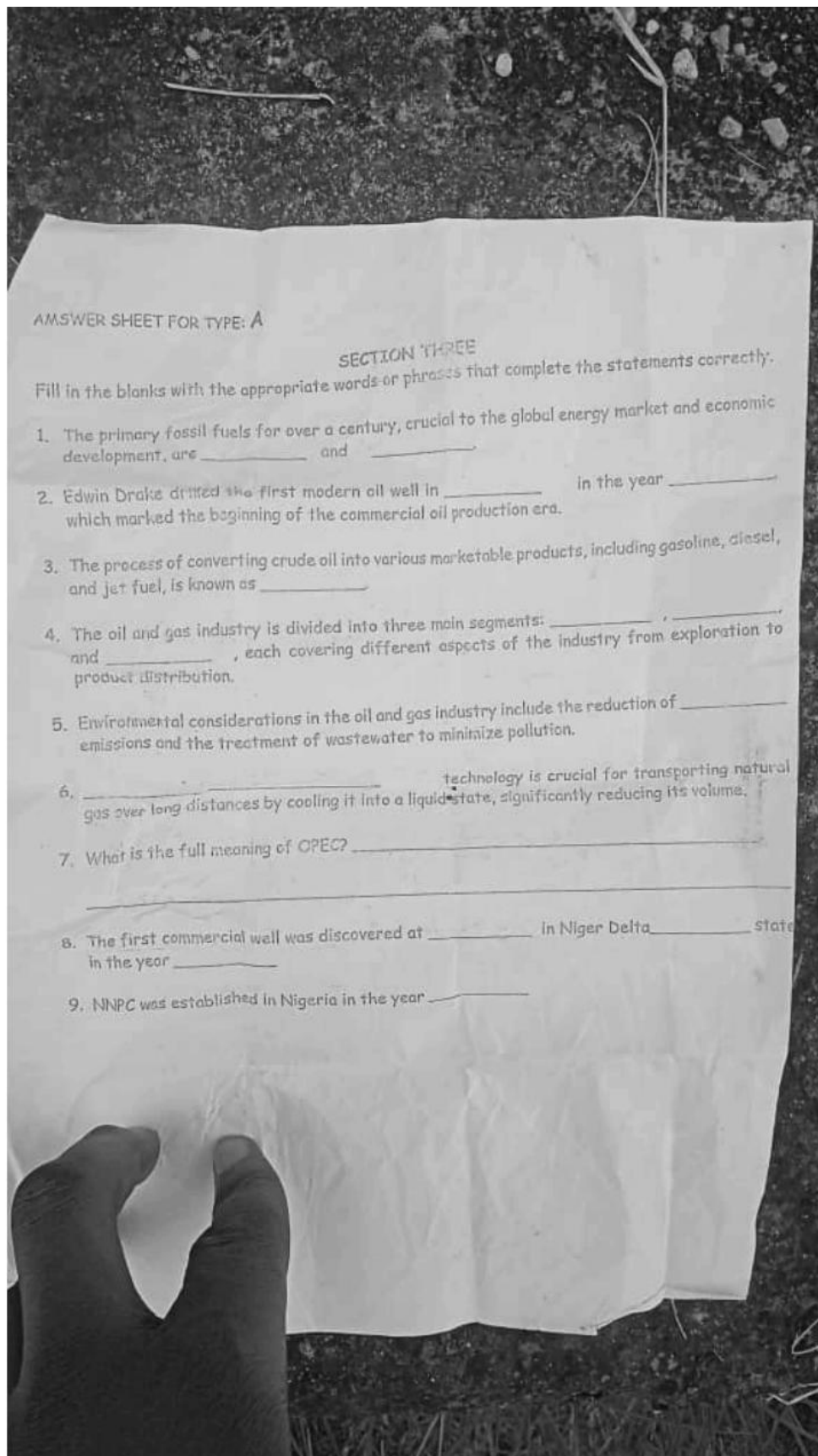
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Paper 4 of 5 — PNG 310

Level 400 • First Semester • PastQuestion

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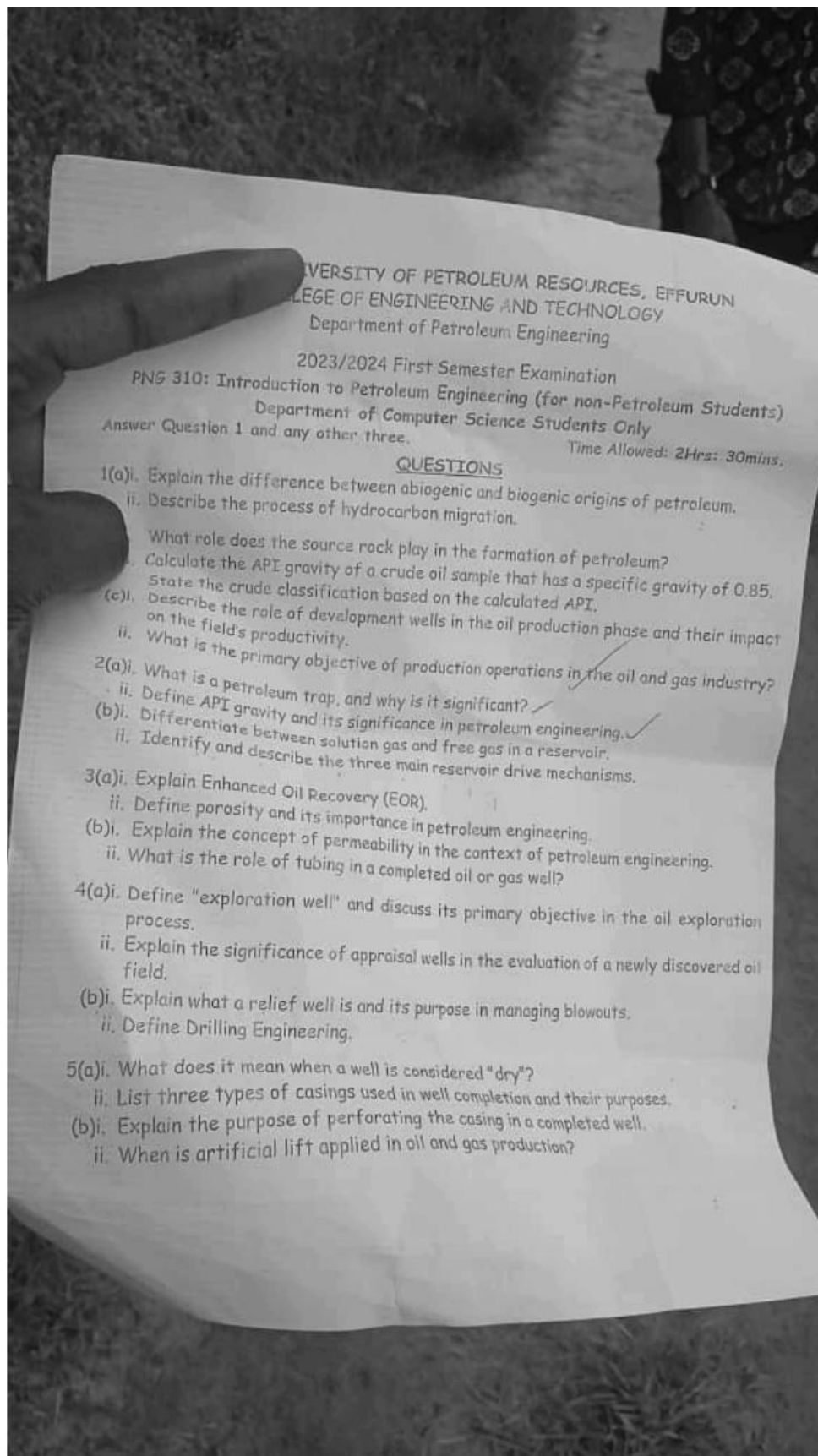


ANSWER SHEET FOR TYPE: A

SECTION THREE

Fill in the blanks with the appropriate words or phrases that complete the statements correctly.

1. The primary fossil fuels for over a century, crucial to the global energy market and economic development, are _____ and _____.
2. Edwin Drake drilled the first modern oil well in _____ in the year _____ which marked the beginning of the commercial oil production era.
3. The process of converting crude oil into various marketable products, including gasoline, diesel, and jet fuel, is known as _____.
4. The oil and gas industry is divided into three main segments: _____, _____, and _____, each covering different aspects of the industry from exploration to product distribution.
5. Environmental considerations in the oil and gas industry include the reduction of _____ emissions and the treatment of wastewater to minimize pollution.
6. _____ technology is crucial for transporting natural gas over long distances by cooling it into a liquid state, significantly reducing its volume.
7. What is the full meaning of OPEC? _____
8. The first commercial well was discovered at _____ in Niger Delta _____ state in the year _____.
9. NNPC was established in Nigeria in the year _____.



Paper 5 of 5 — PNG 310 handout

Level 400 • First Semester • Handout

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